Easy Python Notes: Algorithm, Pseudocode & Shift Conflicts

One-page revision sheet • simple explanations and an example

1. Common Algorithm Structure

Algorithm means the step-by-step plan for solving a problem.

1. **Input:** collect the required values.

2. **Initialize:** create variables/lists and set starting totals to 0.

3. **Process:** use operators, conditions (`if`) and loops (`for`/`while`).

4. **Update:** calculate or change totals/results.

5. **Output:** display the final answer and stop.

2. Pseudocode Template

Pseudocode is a plain-language outline of code. It explains the logic without needing exact Python syntax.

```
START
INPUT required values
INITIALIZE variables / lists / tuples
FOR each item
  CHECK conditions and calculate
  UPDATE totals / results
END FOR
DISPLAY final result
STOP
```

3. Case Study: Employee Shift Conflict Detection

Goal: find employees whose work shifts overlap. A shift can be stored as a tuple such as `(employee, start, end)`.

Steps

- Store each shift as a tuple.
- Compare each pair of shifts.
- Two shifts overlap if `start1 < end2` AND `start2 < end1`.
- If both conditions are true, report a conflict.

Example

Asha: 09:00–13:00
Ravi: 12:00–16:00
These overlap from 12:00 to 13:00, so report a conflict.

4. Python ideas to remember

- **Tuple:** stores related values together; usually cannot be changed.
- **Loop:** repeats steps for many records.
- **Condition:** makes a decision using True/False.
- **Operator:** compares values, e.g. <, >, and.

Exam tip: Write the input, processing steps, condition, and output in that order.

Remember: Input → Process → Check rules → Update result → Output